Software Design Document

Document Number: 03

Date: 22 March 2026

Allies Connect

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Allies Way

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1. INTRODUCTION

1.1 Purpose

This software design document describes the architecture and system design of Allies Connect, a system built for Allies Way (alliesway.org), an organization that restores dignity to families by addressing food insecurity, housing instability, and crisis-related hardship through compassionate service and innovative community support.

It is written primarily for the technical team that will be responsible for deploying, maintaining, and running the web-based application after the code has been submitted to the organization.

1.2 Scope

Allies Connect is intended to be a service that connects Georgians with local community not-for-profit organizations. In an ideal use case any Georgian can open the service and, without creating an account, open up a map that will show them local nonprofits along with their business information so users can easily reach out on their own behalf or on behalf of a client to request aid or provide volunteer services. Allies Connect also provides opportunities for community resource managers to post information about events they will be hosting. Users will be able to navigate through these events by date, location, and other filters to find events that might meet their needs.

Allies Connect also will provide opportunities for volunteers to find community resources and events where they can register to volunteer their time to help. Resources can post events with volunteer opportunities, post volunteer-only events (like a trash-cleanup day), and volunteers will be able to register with community resources to fill shifts. Additionally, Allies Connect will provide reporting services that will assist with the administrative needs of the community resource managers and volunteers by providing information to track event attendance and handle volunteer and resource listing management.

Ultimately, Allies Connect is intended to be a one-stop shop for all the needs of Georgian nonprofits, and the citizens that utilize their services. By consolidating all of this information and management into a single service, users, resource providers, and volunteers should all be able to utilize one single service to fulfill all of their needs. This will hopefully reduce the problem of non-profit service information being scattered across social media, phone networks, and individual websites. When users only have to go check one place to find everything they need, it will be easier for them to find the resources that will best suit their needs instead of tracking them down all over the internet.

1.3 Overview

This document contains descriptions of the system that underpins Allies Connect. It includes a system overview, a review of the system architecture, detailed design information, database design, and user interface/user experience (UI/UX) information.

2. SYSTEM OVERVIEW

The Allies Way system is a web-based application. It is optimized for small screens to accommodate users with older mobile devices. It has UI layouts that can work on the smaller form factor of a phone but which work better on larger devices (tablets, laptops, or monitors), such as the administrative functionality for organization representatives and for Allies Way administrators.

The system's design focuses around resolving three core needs in the nonprofit space:

- There are people that need assistance from nonprofits, but don't know where to find the resources that they need. Sorting through Google results can be difficult, especially if the user doesn't know what kind of help they should be looking for.
- Trying to find information about events that are hosted by local nonprofits can be difficult for one person. Local nonprofits scatter their information across different social media pages, physical advertising, and phone chains. Most nonprofits do not have the budget or resources to hire a marketing manager or dedicated person to manage all of their events.
- Many local Georgian nonprofits require the assistance of volunteers. There are volunteers in the community willing to provide help, but do not always know which nonprofits need their help or when they might need their help.

To help resolve some of these issues, the Allies Connect service offers several systems to help connect Georgians with the resources they need. The first of these is a local resource map. This map uses the user's current location to display community resources near them on an interactive map. The user can filter these options based on the kind of support that they need such as housing assistance or food security assistance. The user can tap on the nodes that show up near them which provide additional detail about the resources like their hours, their eligibility requirements, and their contact information. Leaders of local community groups can sign up for Allies Connect to post their resources to the service, and manage their public information.

Ailles Connect also provides an event finding system to help nonprofits promote all their events in a single place. On the events page users can sort by date, their type of events, and other filters to find events in their area for the specific services they need. These events can support uploading flyers for the event, and even take sign ups for volunteers to assist with running the event. resource providers can host new events from their provider dashboard pages. The event finding service and the local resource map can be used by anyone anywhere without needing a user account.

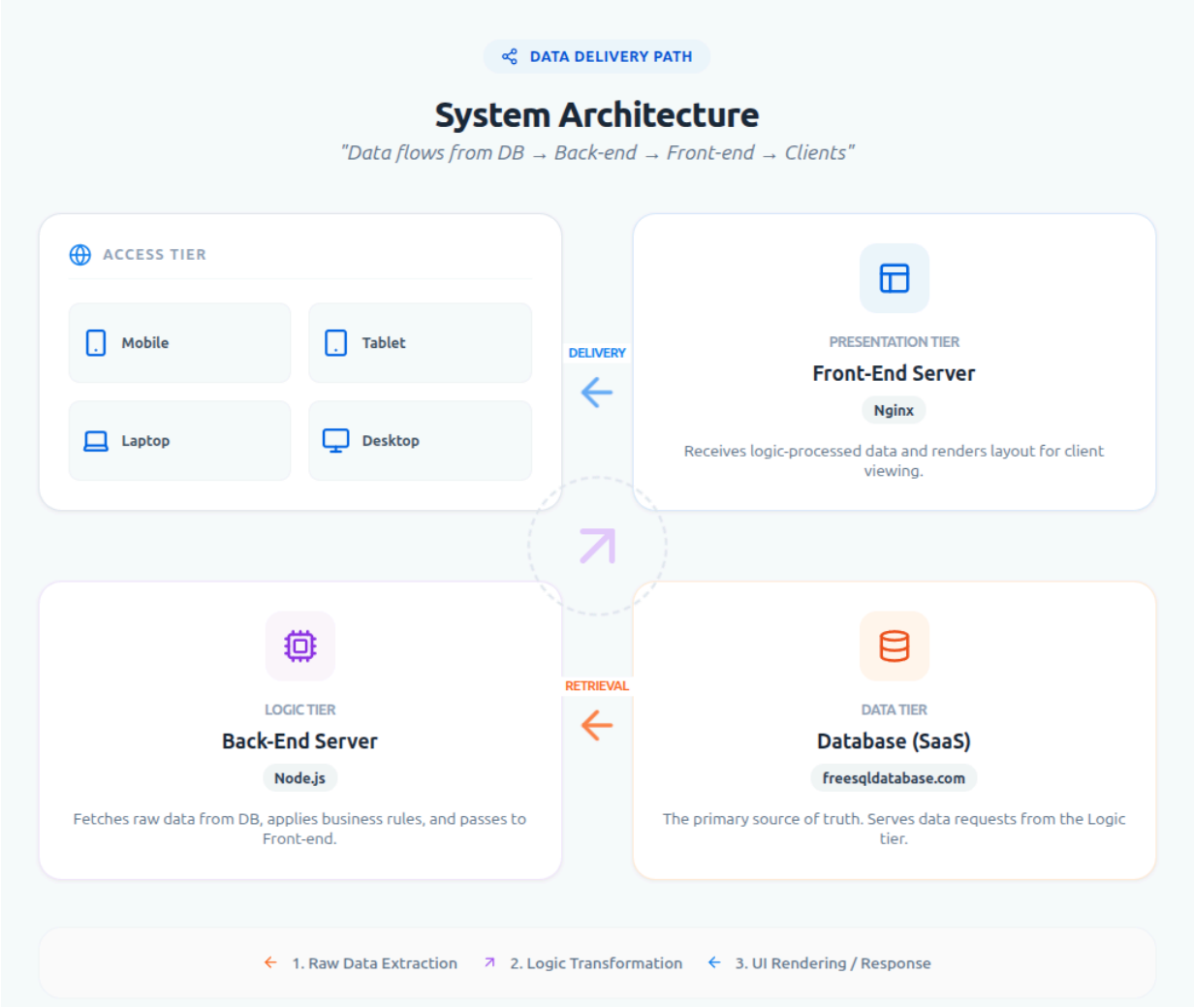
Finally, Allies Connect provides a volunteer management system. This system allows volunteers to sign up for events, and register with local nonprofits to be available to work shifts at those services. Users can sign up as volunteers which gives them access to their volunteer management board where they can select which local nonprofits they have signed up to volunteer with, as well as the events that they have promised to attend as a volunteer. Volunteer users also will see specific volunteer only events that are made available by nonprofits, such as trash pickup days or distribution events.

3. SYSTEM ARCHITECTURE

3.1 Architectural Design

This application follows the standard three-tier design with a database on one tier, a back-end on another, and the front-end on a third tier. The database is hosted and maintained by a software-as-a-service (SaaS) provider called freesqldatabase.com. The back-end server will run on Node. The front-end content will be served by a Nginx implementation.

This separation of concerns ensures that the data integrity is maintained by the database server, the business rules are handled through the back-end server, and the layout and interaction elements are managed by the front-end server.



3.2 Design Rationale

This architecture was selected because of its ubiquity and the suggestion in the original requirements document that the React framework might be a good choice. Reach works well with a three-component system.

4. DETAILED DESIGN

The following sections provide details about the design of the system and the reasons why those decisions were made.

4.1 Access Levels

The design has four different access levels:

- Public user—users with this level of access:
    - Do not need to be logged in
    - Can view lists of events
    - Can view lists of organizations
    - Can view details about specific organizations
    - Can view lists of volunteer opportunities
    - Can view maps that show events and volunteer opportunities
    - Can filter the pins that show up on maps
    - Can sign up to indicate plans to attend an event
- Volunteer—users with this level of access:
    - Must be logged in
    - Can view lists of volunteer opportunities
    - Can sign up to volunteer for events
    - Can update or cancel their volunteer commitments
- Organization representative—users with this level of access:
    - Must be logged in
    - Can view the status of their organization's acceptance
    - Can post/update/delete volunteer opportunities
    - Can post/update/delete events
    - Can add/update/delete information about their organization
    - Can download records about volunteers
    - Can download records about event attendance
- Allies Way administrator (logged in)—users with this level of access:
    - Must be logged in
    - Can create/update/delete other admin accounts (except for the main admin account, which cannot be deleted)
    - Can edit/delete information about an organization
    - Can edit/delete information about a volunteer
    - Can add/edit/delete information about a volunteer commitments
    - Can approve a pending organization record
    - Can add/edit/delete an event
    - Can download records about volunteers
    - Can download records about event attendance

4.2 Public Resource Finder Design

Providing a visual representation of the location of organizations and events is important for individuals who may have limited transportation options. We are including a map representation of the resource location data to enable users to identify the opportunities that are within their

transportation range. The map has the ability to filter based on opportunity type (food insecurity, legal support, and so on) and distance (x miles from the user's location).

A stretch opportunity for future enhancements could include an overlay that shows public transit options (bus, subway/rail, or even ferry if the app is extended to other states).

4.3 Event Directory Design

The event directory contains both static and dynamic information about an event. The static information, applied to all events, includes such information as street address and geolocation. Rather than requiring the system to be updated every time a new field is required, the system enables those creating events to define additional fields specific to their event.

The display of the information for the events directory is covered in the Human Interface Design sections.

4.4 Volunteer Management System Design

The volunteer management system needs to support the needs of organizations to find and assign volunteers to time slots and of volunteers to find opportunities and sign up for those shifts. As with the event directory design, the volunteer management system enables organizations to use the standard fields for sign-ups along with custom, per-opportunity fields. For example, if an event needs to know if volunteers are bringing their own gardening gloves, the organization can ask that in a custom question on the sign-up form.

Additionally, organizations need a simple way of accessing and downloading volunteer information for offline use. We are providing a bulk download option to download the content as a comma-separated value (CSV) file, which can be imported into Google Sheets, Microsoft Excel, and other spreadsheet applications.

4.5 Organization Portal Design

The organization portal provides access appropriate to the individual account that is logging in. It includes the following functions:

- **Account creation**—volunteers can create their own accounts,while organizations can create an account after their organization has been approved, and admin accounts can be created by other admin accounts.
- **Password reset**—All account types can reset and update their own passwords and email addresses.
- **Login**—An individual may have reason to access the site based on all four of the user levels. They may visit the site without logging in to see general information. They may be an Allies Way administrator who is also the organization representative for one or more organizations other than Allies Way, and they may be a volunteer at various events. Because of this, the login functionality prompts the user to specify how they want to log in. Their choice will limit them to the functionality that corresponds with that access level.

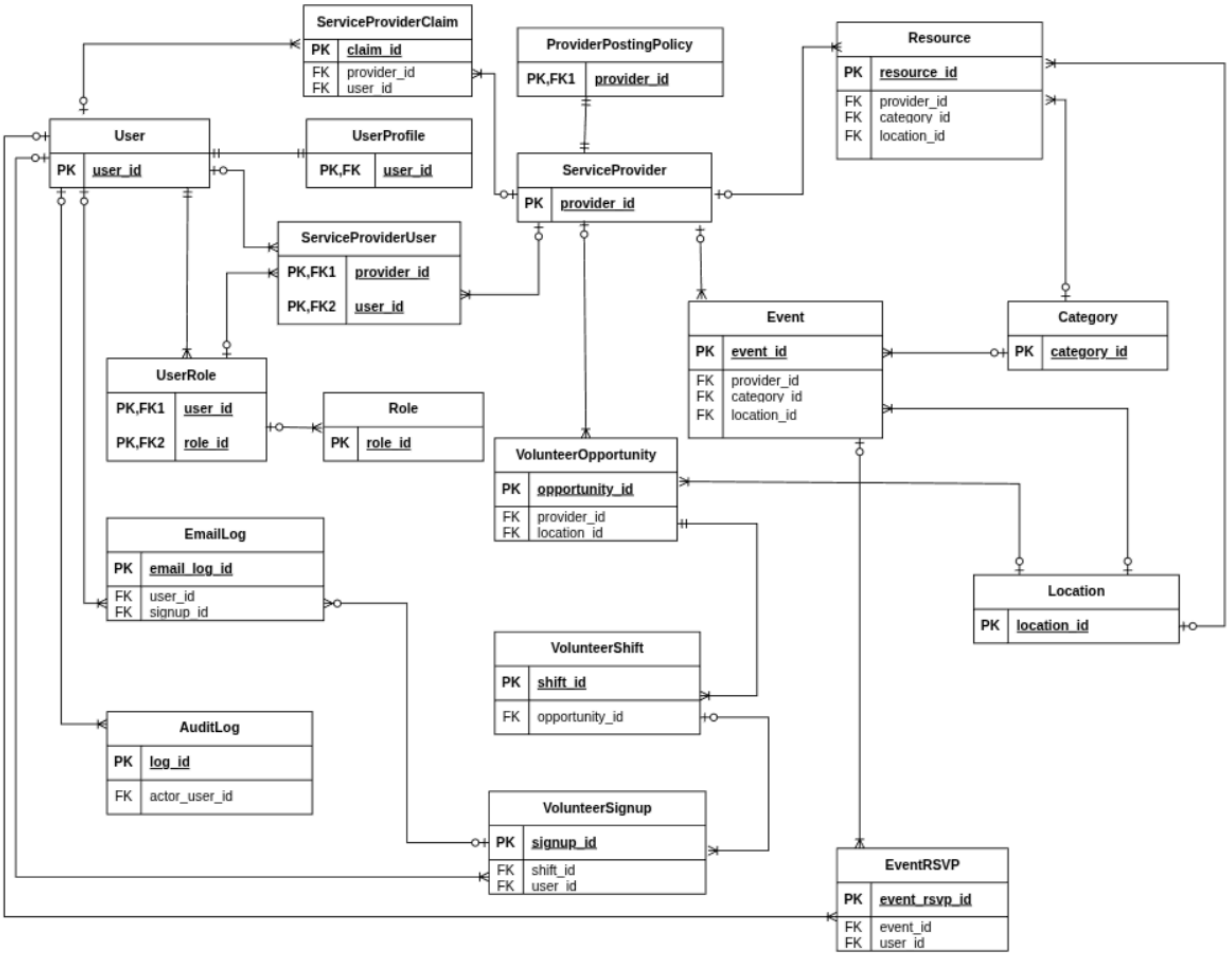
4.6 Administrator Console Design

The administrator console is the primary user interface for Allies Way administrators to manage the site. From this dashboard, they can:

- Approve organizations that have added themselves to the system
- Add, update, and delete information about:
    - Volunteers and their self-assigned shifts
    - Organizations and their events
    - Other administrators
- Administer volunteer, organization representative, and other administrator accounts
- Review metrics for site access
- Review logs of activity on the site

5. DATA (DATABASE) DESIGN

The following is an entity-relationship diagram (ERD) for the database that underpins the Allies Connect web application:



The following sections provide details about the entities in this structure.

5.1 Core Functional Modules[1]

5.1.1 Identity & Access Management (IAM)

The system employs a Role-Based Access Control (RBAC) model.

`User` & `UserProfile`: Separates sensitive authentication data (hashes, status) from descriptive profile data (names, contact info).

`Role` & `UserRole`: Supports a many-to-many relationship allowing users to hold multiple roles

[1] Sections 5.1 and 5.2 and their subsections used Gemini AI to help with identifying the database implementation aspects from the schema structure. It's not a wholesale copy, but there's a fair bit of the AI-generated content still in it.

(e.g., a user can be both a *volunteer* and a *provider*; see 4.5 Organization Portal Design).

Supported Roles: User, Volunteer, Provider, Admin.

5.1.2 Service Provider Infrastructure

This module manages organizations that provide community resources and events.

`ServiceProvider`: The central entity for organizations, tracking mission, operating hours, and status.

`ServiceProviderUser` & `ServiceProviderClaim`: Manages the relationship between individual users and organizations, including a "claiming" workflow for verification (EIN or manual).

`Location`: A centralized table for geospatial data (latitude/longitude) and physical addresses, reused across providers, resources, and events.

5.1.3 Content & Engagement (Resources & Events)

This module handles the primary value proposition of the platform.

`Category`: A categorization system for both events and resources.

`Resource`: Persistent services (e.g., food banks, clinics).

`Event`: Time-bound occurrences (e.g., workshops, clinics) with capacity management.

`EventRSVP`: Tracks user commitment to events, preventing over-capacity registrations.

5.1.4 Volunteer Management

A sub-system for organizing community labor.

`VolunteerOpportunity`: The high-level “job” that can be linked to a specific `Event` or `Resource`.

`VolunteerShift`: Granular time slots for an opportunity, each with its own capacity.

`VolunteerSignup`: Tracks volunteer registration and prevents double-booking via unique constraints.

5.1.5 System Operations & Observability

`EmailLog`: Provides a queue-based system for transactional emails (confirmations, 24h reminders) for notifying volunteers, event coordinators, and others of upcoming events or changes in plans.

`AuditLog`: Captures who did what and when across the system for security and troubleshooting.

5.2. Data Integrity & Logic Implementation

5.2.1 Referential Integrity

Cascading Deletes: Used extensively on child tables (e.g., `VolunteerSignup` deleted if `VolunteerShift` is removed) to prevent orphaned records.

Restrict Constraints: Used on critical lookup data (e.g., `Role`, `Category`) to prevent the deletion of types currently in use.

5.2.2 Security Features

Status Enumerations: The use of ENUM for `User.status` (*active*, *suspended*) and `ServiceProvider.status` (*pending*, *active*) allows for immediate, database-level locking of accounts or visibility of services.

Unique Constraints: Strictly enforced on emails, role names, and RSVPs to prevent duplicate data entry.

5.2.3 Performance Considerations

Normalization: By separating `Location` and `UserProfile` from the core `User` and `ServiceProvider` tables, the system maintains small row sizes for high-frequency authentication queries.

Indexing: Unique keys on email and foreign key columns ensure high-performance lookups during join operations.

6. HUMAN INTERFACE DESIGN

6.1 UI design

For the development of the Allies Connect User Interface, we tried to follow a very simplistic design language that's reminiscent of other modern websites. We used a limited color palette to avoid clutter on the screen, and utilized colors that were already included in the sponsor's company website. Allies Connect is not intended to be a for-profit product, or to keep people on the website for an extended period of time to generate ad revenue like a commercial website might. As a result, we concluded that it was more beneficial for the users and the sponsor if the site prioritized functionality over form. To this end, we did not include extraneous pictures or text, and tried to leave as little room for ambiguity as possible.

A core component of our design also focused around ensuring that the Allies Connect was a mobile-focused web service. Our layouts focused on a clean, easily scalable UI with a mobile-first design philosophy. Content was designed to easily stack and reorder for smaller screens, even for the pages that involve more clerical work like the admin, sponsor, and volunteer account dashboards. We wanted all aspects of the site to be accessible and functional, even on smaller devices. System navigation flows were also developed with a mobile first solution in mind, prioritizing flows that were easy to step through one-by-one without a lot of extra page loading and tapping around.

6.2 UX design

When considering the user experience design for the Allies Connect service our highest priority was ensuring the accessibility of the site for users. Our logic is simple, there is no common denominator for users that will be using the site. Anyone anywhere can fall in hard times at any time, which means that our site may be used by anyone anywhere at any time. The material conditions that could lead someone to needing to rely on a non-profit organization for food security or housing assistance are the same kind of conditions that could lead to difficulty accessing or using the internet. Therefore, it was our highest priority to consider the possibility of user disability when designing our site.

To that end we utilized minimal text and large easily visible buttons with a contrasting color scheme to ensure that text was visible and easy to read. We also utilized screen readers and variable width screen sizes to ensure that the content was usable by visually impaired users, as well as users that may not have the most recent smartphones available. The core features of the site (resource location, event finding, and volunteer signups) can be mostly navigated without the use of the keyboard, but are easily navigable by using keyboard controls. The visual hierarchy is clear on both mobile and desktop versions of the site, and the system has been designed in a lightweight way to feature snappy response times.